

Supplementary Material for *Aligning Multi-Trajectory Supervision with Policy Optimization for VLA Driving*

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Abstract

This supplement specifies the three stages of Aligned Multi-Trajectory Policy Training (AMPT), records implementation and evaluation provenance, and expands the supervision, credit-assignment, policy-refinement, and failure-recovery analyses. Superscripts R, S, and P denote paper-reported, scene-level reproduced, and proposed-but-unrun reproduction settings, respectively. This distinction prevents a proposed control from being mistaken for an observed result. Unless stated otherwise, evaluation emits one trajectory and performs neither test-time scoring nor candidate reranking.

A Additional Method Details

This appendix makes explicit the contracts that connect the three training stages. This is important because the central claim of AMPT is not that a trajectory with a high offline score is automatically a useful target, but that supervision, rollout credit, and policy-relative refinement must agree on what the current policy can safely learn. We use three provenance labels throughout. *Paper-reported* denotes a value stated in the main paper, *scene-level reproduced* denotes a value recomputed from an archived per-scene evaluator export, and *proposed reproduction* denotes a fully specified setting for rerunning an experiment whose resolved training manifest was not retained. The last category is a protocol, not an observed result.

A.1 Policy-Compatible Multi-Trajectory Supervision

Candidate construction and source balancing. For a scene s , let τ_s^0 be the logged trajectory and let $\mathcal{C}_s = \cup_m \mathcal{C}_s^{(m)}$ be the union of candidates supplied by the configured proposal generators. PC-MTS is source agnostic: a source may be another policy seed, a structured trajectory expander, or a specialist planner, but source identity is retained in the manifest and geometric near-duplicates are merged before scoring. The logged target is immutable and is never removed. A final Stage-1 source-count manifest was not found. To make the controlled reconstruction executable without presenting invented counts as measurements, our *proposed reproduction* caps

every non-GT source at eight candidates per scene before filtering. When comparing Score, Pareto, and PC-MTS, we use the same source cap and retain the same number of non-GT candidates by taking the best valid items under the corresponding rule. Thus a difference cannot be attributed to exposing one variant to a larger target set.

Quality, feasibility, and Pareto filtering. Every complete trajectory is scored by the same offline evaluator used for training diagnostics. We first require hard feasibility, then a sufficient aggregate score, and only then form the scene-wise component Pareto front. A candidate a dominates b only if it is no worse on every active Pareto objective and is strictly better on at least one. This order matters: a high progress value cannot compensate for collision or leaving the drivable area. The paper does not archive the numerical Stage-1 aggregate cutoff. In the *proposed reproduction*, a training-split quantile is selected once so that comparison variants are matched to the minimum retained count, followed by a source-balanced top- k tie break. The sealed cutoff is applied identically to all splits and is never used to synthesize a reported score.

The benchmark-specific partition is summarized below. On NAVSIM v1, NC and DAC are hard gates; DDC is protected; and EP, TTC, and comfort are optimized inside the admissible region. On NAVSIM v2, NC, DAC, DDC, and TLC are the multiplicative compliance gates. The historical metric adapter uses EP, TTC, and a composite $Q = (\text{LK} + \text{HC} + \text{EC})/3$ as its three continuous Pareto axes; the official EPDMS report still exposes LK, HC, and EC separately. For binary scorer outputs the *proposed reproduction* requires a value of one up to 10^{-6} . The archived safety contract uses reference-relative tolerances of 0.01 for DDC/comfort and 0.02 for TTC/EP; TLC remains a hard v2 gate. These are historical implementation values, not a claim about the unresolved configuration of the paper checkpoint.

Policy compatibility. Let $\mathcal{R}_s = \{\rho_j\}_{j=1}^M$ be rollouts sampled from the frozen GT-only policy under the same command as the candidate. We standardize waypoint displacement with per-timestep, command-conditioned robust scales fitted on the calibration split, and write the resulting trajectory distance as $d(\tau, \rho)$. The archived implementation additionally exposes final displacement,

Table 1: Candidate-source and audit budgets for PC-MTS on the NAVSIM training universe. Counts tagged proposed are reproduction settings rather than observations; higher/lower is not applicable.

source	role	per_scene_budget	scene_universe	retention_or_use	status
Logged trajectory	immutable fallback and retention target	1	103,288 expected navtrain scenes	always retained	paper-described / historical contract
Frozen-policy rollout bank	compatibility support estimate	16	103,288-scene reproduction universe	distance calibration only	proposed reproduction setting
Multi-source candidate proposals	quality and Pareto targets	8 per source	103,288-scene reproduction universe	subject to quality/Pareto/compatibility gates	proposed reproduction budget
Correlated local perturbations	local feasibility audit	8 per retained proposal	same candidate scenes	pass-rate threshold 0.75	proposed reproduction setting

Table 2: NAVSIM v1/v2 metric partition used by PC-MTS and FF-PGRPO. Hard feasibility is tested before protected non-regression and Pareto quality.

benchmark	hard_feasibility	protected_guards	pareto_quality	aggregate
NAVSIM v1	NC, DAC	DDC; non-regression in TTC and feasibility	EP, TTC, C	PDMS
NAVSIM v2	NC, DAC, DDC, TLC	DDC/TLC feasibility; non-regression in TTC	EP, TTC, Q=(LK+HC+EC)/3	EPDMS

heading, and velocity diagnostics, but the proposed controlled comparison changes only the normalized waypoint statistic. The compatibility statistic is

$$d_{PC}(\tau, \mathcal{R}_s) = \frac{1}{K} \sum_{\rho \in \text{NN}_K(\tau; \mathcal{R}_s)} d(\tau, \rho). \quad (1)$$

The threshold q_c is the command-specific quantile of leave-one-out policy rollout distances on calibration scenes. The archived code proves leave-one-out calibration within each policy cloud, but its retained manifest does not prove driving-log-level independence. The proposed reconstruction therefore enforces log-disjoint calibration and query scenes. The paper states the general K -nearest statistic but does not report M , K , or q_c . The archived implementation realizes the valid $K = 1$ special case. Our *proposed reproduction* uses $M = 16$, $K = 3$, and q_c at the 95th percentile, with command groups merged into a global fallback only when a command group has fewer than 16 calibration observations. This choice is also the center point of the supplied sensitivity grid, rather than a retrospectively selected result.

Local feasibility tube. Compatibility to one rollout neighborhood does not by itself establish that a target is robust to small prediction errors. For every candidate that passes the previous gates, PC-MTS adds correlated perturbations to (x, y, ψ) and the implied velocity sequence. The perturbations are sampled from normalized policy residuals, smoothed over time, and projected back to the command-consistent trajectory representation before exact re-evaluation. This changes whole trajectory segments rather than independently jittering waypoints. The archived implementation supports 16 or 32 perturbations; the *proposed reproduction* uses eight lower-cost perturbations, preserves the empirical temporal covariance of policy residuals, and accepts a candidate when at least six of eight perturbations pass the hard feasibility gates and the protected guards. The P1 configuration evaluates 4/8/16 perturbations and does not tune the threshold on the test split.

Target sampling and fallback. After filtering, the training set for scene s is $\mathcal{T}_s = \{\tau_s^0\} \cup \mathcal{A}_s$, where $\mathcal{A}_s$

contains accepted candidates. A training update draws exactly one target for the scene, so scenes with many proposals do not receive larger gradients. The *proposed reproduction* draws the logged target with probability 0.5 and otherwise samples uniformly from $\mathcal{A}_s$; if $\mathcal{A}_s$ is empty, it deterministically uses τ_s^0 . Candidate identity is sampled before the diffusion timestep, so all noisy versions in that update correspond to a single coherent target. Fine-tuning uses the same denoising objective as GT-only training and produces π_{initial} .

Implementation correspondence. The source-balanced builder and immutable-GT path are implemented at 41a8613:navsim/agents/recogdrive/curriculum/builder.py and 41a8613:navsim/agents/recogdrive/curriculum/proposals.py. Physical distance and calibrated reachability are at 41a8613:navsim/agents/recogdrive/curriculum/reachability.py; the feasibility and correlated-perturbation checks are at 41a8613:navsim/agents/recogdrive/curriculum/safety.py and 41a8613:navsim/agents/recogdrive/curriculum/robustness.py. These are historical mechanism-level implementations; the audit does not claim that this commit is a complete hash-identical release of the final checkpoint.

A.2 Feasibility-First Pareto GRPO

A coherent scene reference. FF-PGRPO compares a rollout with one executable reference trajectory, never with an envelope assembled from different trajectories. The ordered source set is: the logged trajectory, a retained PC-MTS candidate, and the current single-trajectory prediction. The paper-level rule selects the highest-scoring feasible row. The audited historical selector makes this precise by ranking validity, hard feasibility, reference guard, robustness, core eligibility, PDMS, distance to the current prediction, and source priority lexicographically. Thus no higher score can bypass an earlier safety/eligibility tier. If none is feasible, the logged trajectory is retained only as a weak recovery target when it passes the evaluator’s basic physical checks. Otherwise no recovery target is used and every sampled rollout receives non-positive credit. A component-wise envelope is invalid because its NC, progress, and TTC values may come from mutually incompatible plans; positive credit relative to such a synthetic point would have no executable safety witness.

Group states and admissibility. For each scene the paper reports a group of $G = 16$ policy rollouts. We assign

the group to one of three states. An *all-feasible* group contains only rollouts that satisfy hard feasibility; its quality statistic normalizes the continuous score components because the binary gates are constant. A *mixed* group contains both feasible and infeasible rollouts; feasibility is applied first and feasible trajectories must additionally pass the coherent reference guards. An *all-infeasible* group contains no feasible rollout; it is not allowed to create a positive advantage by merely ranking unsafe samples against one another.

For NAVSIM v1, admissibility requires NC and DAC and protects DDC, TTC, and an EP floor relative to the coherent reference. NAVSIM v2 additionally treats DDC and TLC as hard compliance checks. The archived safety-contract defaults use $\text{DDC}(\tau) \geq \text{DDC}(r) - 0.01$, $\text{TTC}(\tau) \geq \text{TTC}(r) - 0.02$, and $\text{EP}(\tau) \geq \text{EP}(r) - 0.02$. The joint EP-TTC guard prevents the optimizer from receiving positive credit for increasing safety merely by stopping, or for increasing progress by collapsing the collision margin.

Credit assignment. Within a group that contains a feasible rollout, let $\hat{A}_i$ be a quality-shaped, group-relative score and let P_i indicate membership in the admissible Pareto front. FF-PGRPO assigns

$$A_i = \begin{cases} \max(\hat{A}_i, 0), & \text{feasible, guard-passing,} \\ & \text{and Pareto-positive} \\ \min(\hat{A}_i, 0), & \text{feasible but not} \\ & \text{admissible Pareto} \\ -\lambda_{\text{unsafe}} + \min(\hat{A}_i, 0), & \text{infeasible.} \end{cases} \quad (2)$$

Thus aggregate quality controls magnitude, whereas feasibility, a coherent reference, and non-domination control the sign. For mixed groups the quality term also includes the EP-floor and joint EP-TTC penalties. The historical implementation preserves signs with RMS scaling rather than mean centering and uses an archived default clip of $[-3, 3]$; the *proposed reproduction* uses the predeclared sensitivity setting $[-5, 5]$.

For an all-infeasible group, violation severity is the normalized sum of hard gate deficits and protected-metric deficits. Credit is the non-positive within-group rank of negative severity, shifted by the unsafe penalty. The group loss is downweighted to 0.25; when a valid recovery reference exists, a diffusion recovery term with coefficient 0.25 is added. The proposed regularization uses KL weight 0.02 and anneals the BC coefficient linearly from 0.10 to 0.05 over the first five epochs. These numerical values are historical implementation defaults and proposed rerun values, not recovered resolved parameters for the paper’s final FF-PGRPO run.

Diffusion credit and inference. One trajectory-level advantage is broadcast to every transition in that trajectory’s denoising chain. The audited planner uses a discounted transition-wise mean, while the paper objective writes the chain likelihood in accumulated form; the proposed reconstruction preserves the planner reduction and logs it in the resolved manifest. This avoids assigning

mutually inconsistent metric credit to intermediate noisy states that are not executable plans. At inference, AMPT runs one denoising chain and emits one trajectory. Group sampling, the Pareto computation, the reference selector, the offline evaluator, and candidate reranking are absent from test time.

Implementation correspondence. Group classification, Pareto gating, violation ranking, sign-preserving normalization, and clipping are at `41a8613:navsim/agents/recogdrive/stage3_ff_pareto.py`. The coherent-row API is at `41a8613:navsim/agents/recogdrive/stage3_reference_selector.py`, and positive-credit assertions are at `41a8613:navsim/agents/recogdrive/stage3_positive_credit_contract.py`. The archived tensor layout is anchor-local; the paper-level protocol is the union of 16 scene rollouts. Because a final resolved layout manifest was not found, we do not infer the number of anchors from a class default.

A.3 Adaptive Pareto-Guided Policy Refinement

Dynamic multi-source pool. At APR round r , the current policy π_r supplies one local prediction for every scene. Candidate teachers come from historical policy checkpoints, independent GRPO seeds, structured trajectory expansion, and safety-, progress-, and regularity-oriented specialists. Source labels survive deduplication so per-source acceptance can be audited, but final three-round source proportions were not archived. The audited initial construction keeps its observed 25,402/13,859 source mixture. The proposed source-ablation reports each source separately instead of claiming an unrecorded balancing rule.

Teacher audit. For a policy prediction p_s and candidate teacher t , APR first requires feasibility and an aggregate margin $R(t) - R(p_s) \geq m_R$. It then applies absolute and reference-relative protected guards. Behavioral distance, curvature, and jerk rank otherwise admissible teachers; they are preferences, not claimed historical hard thresholds. In the audited APR branch, $m_R = 0$ with a strict positive comparison, DDC has absolute floor 0.99 and maximum drop 0.01, TTC has absolute floor 0.95 and maximum drop 0.02, and the jerk coefficient 0.005 is a soft training regularizer. At the interpolation audit, the proposed reconstruction additionally applies the command-conditioned 95th-percentile compatibility calibration from PC-MTS.

Local interpolation and exact re-evaluation. An accepted teacher is not copied directly. APR evaluates

$$\tau_{s,\alpha} = p_s + \alpha(t - p_s) \quad (3)$$

for a descending list of interpolation coefficients. The first (largest) coefficient that remains feasible, improves the ag-

aggregate score, preserves protected metrics, and passes the current compatibility test becomes the local target. Every interpolated trajectory is scored anew; endpoint scores are never linearly interpolated. One audited branch used $\alpha = 0.25$. The *proposed reproduction* fixes the descending list to $[1.0, 0.75, 0.50, 0.25]$, which includes that audited setting and tests the teacher endpoint before progressively more conservative steps. If all coefficients fail, the teacher is rejected for the current round rather than kept with a smaller unscored step.

Weighted distillation, retention, and lifecycle. For verified gain $\Delta_s = R(\tau_{s,\alpha}) - R(p_s) > 0$, the target weight is

$$w_s = \min\{5, \exp(\Delta_s/0.05)\}. \quad (4)$$

The temperature 0.05 and clip 5 are present in an audited APR training branch. The same diffusion denoising loss is applied to weighted local targets, while current predictions provide retention targets with audited weight 1.0. At most the top one positive-advantage target is used per scene; a scene without a valid teacher contributes retention only. After training, the candidate checkpoint is promoted only if aggregate validation score increases and none of the protected aggregate components regresses beyond the audit tolerances. The promoted checkpoint becomes the dynamic anchor, its predictions and rollout bank are regenerated, and the complete teacher pool is audited again. A previously accepted teacher is defined as *expired* when it loses its margin or guard, while a previously rejected one is *newly activated* when it passes under the new policy. Historical scripts rebuild round files but do not persist these labels; the proposed pipeline obtains them by joining teacher hashes across consecutive round manifests. The paper reports three rounds. The proposed stop rule is the earlier of three promotions, an empty teacher set, or no checkpoint passing the promotion rule; it does not turn a small unverified score fluctuation into a promotion.

Optional branch consolidation. Safety, progress, and regularity branches may be consolidated after their teacher updates. This is an optional checkpoint operation, not a substitute for teacher auditing. The audited anchor-constrained variant keeps the top 20% salient coordinates of each base-relative task vector, removes coordinates with inconsistent directions, clips every branch displacement to the norm of its accepted branch, and caps total displacement at 1.05 times the largest accepted branch displacement. The anchor is the current promoted policy and is therefore updated across rounds. If no merge candidate passes the same evaluation guard as an ordinary APR checkpoint, the best accepted single branch is retained.

Implementation correspondence. Teacher audits are implemented at `fb96a03:scripts/`

`stage3/build_pdms_strict_teacher_buffer.py` and `fb96a03:scripts/stage3/build_pdms_safety_repair_teacher_buffer.py`. Interpolation and exact rescoring are at `ab9124c:scripts/stage3/build_pdms_blended_teacher_reference.py`, with local safety-tube checks at `ab9124c:scripts/stage3/filter_pdms_teacher_reference_by_safety_tube.py`. Retention schedules are built by `fb96a03:scripts/stage3/build_pdms_teacher_refinement_schedule.py`, and the optional merge is at `64f5531:scripts/stage3/merge_conflict_free_anchored_task_vector.py`. APR is an orchestrated sequence of these audited operations; the repository does not contain a single monolithic `APR.run` entry point.

B Algorithms

The three procedures below are separated deliberately. PC-MTS constructs a supervision distribution before policy optimization; FF-PGRPO assigns credit inside one rollout group; and APR rebuilds policy-relative teachers between training rounds. Combining them into one loop would obscure both their different inputs and the interfaces tested by the implementation. A code comment names the closest audited historical function; “paper contract” marks a step for which no end-to-end final-checkpoint call graph was retained.

Algorithm 1: Policy-Compatible Multi-Trajectory Supervision.

Input: logged scenes $\mathcal{D} = \{(o_s, \tau_s^0, c_s)\}$; candidate sources $\{\mathcal{C}_s^{(m)}\}$; frozen GT policy π_{GT} ; offline evaluator E ; compatibility and robustness configuration ξ .

Output: accepted per-scene target sets $\{\mathcal{T}_s\}$ and the initialized policy π_{initial} .

1. Load source manifests, preserve source identity, form $\mathcal{C}_s = \cup_m \mathcal{C}_s^{(m)}$, merge geometric near-duplicates, and mark τ_s^0 immutable. *Implementation:* `41a8613:navsim/agents/recogdrive/curriculum/proposals.py:load_scene_proposals, deduplicate_proposals;` immutable retention is enforced by `builder.py:finalize_scene_record`.
2. Evaluate every complete candidate with E ; reject candidates that fail hard feasibility, the aggregate-quality threshold, intent checks, or reference-relative protected guards. *Implementation:* `41a8613:navsim/agents/recogdrive/curriculum/builder.py:build_scene_intermediate, safety.py:absolute_feasibility` and `relative_nonregression`.
3. Compute the component-wise Pareto front among survivors and apply the fixed

source/role quota. Do not allow a continuous metric to restore a candidate removed by a hard gate. *Implementation:* component-wise Pareto filtering is the paper contract; the audited tree exposes the later role/quota step at `41a8613:navsim/agents/recogdrive/curriculum/quota_selector.py:role_aware_quota_select`, but no separately named Pareto-front function.

4. Sample the policy rollout cloud $\mathcal{R}_s$ from frozen π_{GT} . Fit robust distance scales and command-specific quantiles using leave-one-out rollout queries. Enforce log-disjoint calibration/query scenes in the proposed rerun; the archived manifest proves leave-one-out only. *Implementation:* `41a8613:navsim/agents/recogdrive/curriculum/builder.py:policy_cloud_for_scene, reachability.py:fit_reachability_calibration`.
5. For each Pareto survivor τ , compute its calibrated nearest-policy distance $d_{\text{PC}}(\tau, \mathcal{R}_s)$ and retain it only when the relevant command threshold is met. The archived branch implements $K = 1$; the proposed rerun sets K through ξ . *Implementation:* `41a8613:navsim/agents/recogdrive/curriculum/reachability.py:physical_distance_components, evaluate_reachability`.
6. Draw temporally correlated residual trajectories around every compatibility-passing candidate, score each perturbation with E , and keep the center candidate only if the required fraction remains feasible and guard-passing. *Implementation:* `41a8613:navsim/agents/recogdrive/curriculum/robustness.py:CorrelatedErrorModel.sample_perturbations, evaluate_robustness_tube`.
7. Set $\mathcal{T}_s = \{\tau_s^0\} \cup \mathcal{A}_s$. If no non-GT candidate survives, set $\mathcal{T}_s = \{\tau_s^0\}$; never borrow a target from another scene. *Implementation:* `41a8613:navsim/agents/recogdrive/curriculum/builder.py:finalize_scene_record`.
8. At each visit to s , sample exactly one member of $\mathcal{T}_s$ according to the fixed GT/candidate mixture and then sample one diffusion timestep. *Implementation:* `41a8613:navsim/agents/recogdrive/stage2_role_sampler.py, stage2_bridge.py`.
9. Fine-tune a copy of π_{GT} with the unchanged diffusion noise-prediction loss and return it as π_{initial} . *Implementation:* `41a8613:navsim/agents/recogdrive/first_drive_v7_planner.py`.

Algorithm 2: Feasibility-First Pareto GRPO Credit Assignment.

Input: scene o_s ; policy π_θ ; the paper’s $G = 16$ sampled denoising chains and final trajectories

$\{(z_i, \tau_i)\}_{i=1}^G$; logged, retained, and current-policy reference candidates; evaluator E ; credit configuration ζ .

Output: trajectory advantages $\{A_i\}$ and the regularized FF-PGRPO loss.

1. Convert exact evaluator outputs for every τ_i to hard-feasibility, continuous-quality, protected-guard, and violation-severity tensors. *Implementation:* `41a8613:navsim/agents/recogdrive/stage3_metric_adapter.py`.
2. Rank complete logged, retained, and current-policy rows lexicographically by validity, hard feasibility, reference guard, robustness, core eligibility, PDMS, distance, and fixed source priority; choose the first row r_s . Never take a different trajectory for each reference component. Record whether a safe recovery reference is available. *Implementation:* `41a8613:navsim/agents/recogdrive/stage3_reference_selector.py:CoherentReferenceSelector.select, gather_reference_rows`.
3. Classify the rollout group as ALL-SAFE, MIXED, or NO-SAFE. For safe rollouts, evaluate coherent-reference guards and admissible Pareto membership. The audited function consumes anchor-local $[B, K, R]$ tensors; the final $K \times R$ layout is unresolved, so the proposed rerun seals $K = 8, R = 2$ to preserve $G = 16$. *Implementation:* `41a8613:navsim/agents/recogdrive/stage3_ff_pareto.py:classify_group_state, anchor_local_pareto`.
4. If the state is ALL-SAFE, form $\hat{A}_i$ from only the continuous score terms. If it is MIXED, form $\hat{A}_i$ from aggregate quality plus EP-floor and joint EP-TTC shaping. Preserve the scene grouping during normalization. *Implementation:* `41a8613:navsim/agents/recogdrive/stage3_ff_pareto.py:FeasibilityFirstCreditAssigner.compute`.
5. For an ALL-SAFE or MIXED group, set positive credit only for safe, guard-passing, Pareto trajectories; clamp safe dominated or guard-failing trajectories to non-positive credit; and apply an additional unsafe penalty to infeasible trajectories. *Implementation:* `41a8613:navsim/agents/recogdrive/stage3_ff_pareto.py:FeasibilityFirstCreditAssigner.compute`.
6. For a NO-SAFE group, rank negative violation severity and assign only non-positive credit. Downweight the group and, if r_s is a valid recovery reference, add weak diffusion supervision toward r_s . *Implementation:* `41a8613:navsim/agents/recogdrive/stage3_ff_pareto.py:FeasibilityFirstCreditAssigner.compute`; the exact final recovery-loss integration is not retained.
7. Apply sign-preserving RMS scaling and symmetric clipping. Assert that no positive-credit

rollout is infeasible, dominated, reference-regressing, or drawn from a no-safe group; log both numerator and denominator of every violation rate. *Implementation:* 41a8613:navsim/agents/recogdrive/stage3_ff_pareto.py: rms_scale_without_mean_centering, stage3_positive_credit_contract.py: assert_positive_credit_contract.

8. Broadcast A_i over the transitions of chain z_i , compute the planner’s discounted transition-wise mean, and optimize it with KL/BC and optional recovery losses. No evaluator or group operation is exported to inference. *Implementation:* the historical transition reduction is in 41a8613:navsim/agents/recogdrive/first_drive_v7_planner.py; the audited commit does not expose a verified end-to-end call from that planner to FeasibilityFirstCreditAssigner.

Algorithm 3: Adaptive Pareto-Guided Policy Refinement.

Input: FF-PGRPO policy π_0 ; multi-source teacher pool $\mathcal{P}$; evaluator E ; interpolation list $\mathcal{L}$; number of rounds $R = 3$; audit and promotion configuration η .

Output: promoted single-trajectory policy π_R and a round-wise teacher lifecycle manifest derived by joining teacher hashes.

1. At round r , run π_r once on every training scene and cache the prediction, score components, trajectory geometry, and rollout-neighborhood calibration as the dynamic anchor. *Implementation:* policy inference and evaluator exports precede fb96a03:scripts/stage3/build_pdms_teacher_refinement_schedule.py: make_refinement_schedule, which consumes rather than creates those cached predictions.
2. Load and deduplicate candidates from historical policies, independent GRPO seeds, structured expansion, and safety/progress/regularity specialists. Retain source and trajectory hashes in every record. *Implementation:* ab9124c:scripts/stage3/build_pdms_external_safety_repair_teacher_buffer.py: choose_external_safety_teachers.
3. Relative to the current prediction, reject a candidate unless it is feasible, clears the score margin, and preserves protected metrics. Rank survivors by verified gain and then by smaller behavioral distance, curvature, and jerk; select at most one teacher for the scene. These last quantities are preferences, not claimed historical hard gates. *Implementation:* fb96a03:scripts/stage3/build_pdms_strict_teacher_buffer.py: _strict_teacher_upgrade, build_teacher_record.
4. For the selected teacher t_s , test $p_s + \alpha(t_s - p_s)$ in descending $\alpha \in \mathcal{L}$. Score every interpolation

with E and choose the largest one that remains feasible, improves score, preserves guards, and remains policy compatible. If none passes, create no teacher target for s . *Implementation:* the audited script evaluates one supplied α per invocation; descending search is the proposed outer orchestration around ab9124c:scripts/stage3/build_pdms_blended_teacher_reference.py: blend_trajectory, _score_one, accept_scored_teacher.

5. Assign each verified target the clipped exponential advantage weight and construct a teacher/retention schedule that records source identity. Scenes without an accepted target contribute only their current prediction as retention. *Implementation:* weighting is in ab9124c:navsim/agents/recogdrive/recogdrive_diffusion_planner.py; retention sampling is in fb96a03:scripts/stage3/build_pdms_teacher_refinement_schedule.py: make_refinement_schedule.
6. Fine-tune the diffusion planner with weighted teacher loss, retention loss, and KL/distillation control. If specialist branches are enabled, train each from the same dynamic anchor and with the same step budget. *Implementation:* ab9124c:scripts/stage3/launch_pdms91_external_teacher_retention_3ep.sh.
7. Optionally consolidate accepted specialist branches using salient, direction-consistent coordinates and an anchor-relative norm bound; also retain every unmerged branch as a promotion candidate. *Implementation:* 64f5531:scripts/stage3/merge_conflict_free_anchored_task_vector.py: conflict_free_anchored_graft.
8. Evaluate all candidates with single-trajectory inference. Promote only a candidate that improves aggregate validation score without violating protected-component tolerances; otherwise keep π_r . Record the input, output, and evaluator hashes. *Implementation:* ba5908d:navsim/planning/script/run_pdm_score_from_submission.py: run_pdm_score for NAVSIM v1 and ab9124c:scripts/evaluation/score_recogdrive_predictions_navsim_v2.py: main for NAVSIM v2.
9. Rebuild the pool relative to the promoted policy. Mark teachers that no longer pass as expired and previously rejected candidates that now pass as newly activated by joining source/trajectory hashes across consecutive manifests. Stop after R rounds or earlier if the teacher set is empty or no checkpoint passes promotion. Return the last promoted policy, not the numerically largest unguarded branch. *Implementation:* the historical scripts rebuild each round but do not emit lifecycle labels; the proposed manifest join supplies them. No monolithic APR entry point is claimed.

C Implementation and Evaluation Protocol

This section separates the protocol that produced the paper’s reported results from settings chosen to make the released reconstruction executable. It also defines the unit of inference and the failure-recovery population; without these distinctions, additional samples, evaluator-guided reranking, or a different 658-scene checkpoint pair could silently change the claim being tested.

C.1 Architecture and Trajectory Interface

AMPT uses the paper-reported InternVL3-2B visual-language backbone and the ReCogDrive conditional diffusion planner. The paper describes multi-view camera features, ego state and motion history, and a discrete high-level command. One audited APR command instead resolves `cam_type=single`; because the exact final checkpoint manifest is incomplete, this is recorded as a protocol discrepancy rather than silently attributed to every reported row. The planning head predicts a four-second ego trajectory as eight poses at 0.5-second spacing, stored as an 8×3 tensor of local-frame $(x, y, \text{heading})$. The VLM context conditions every diffusion step; the supervised objective samples a noisy version of one target trajectory and regresses the corresponding diffusion noise.

The archived agent configuration sets `train_backbone=false` and the planner accepts cached VLM features. Accordingly, the *proposed reproduction* freezes InternVL3-2B and optimizes the ReCogDrive diffusion planner in all four training phases. This makes the supervision variants differ only in target construction and makes the policy-optimization variants differ only in credit and refinement. The archived planner default uses five DDIM denoising steps; the proposed rerun keeps five steps for rollout sampling, log-probability evaluation, and final inference. A random training diffusion timestep is still used for ordinary denoising supervision.

C.2 Stage-Wise Training Protocol

The main paper reports 200 epochs for GT-only and multi-trajectory imitation learning, 16 rollouts per scene and 10 epochs for GRPO, three APR rounds, and eight NVIDIA A800 GPUs. An archived historical Core-Pareto launcher contains 20 epochs; because it is not the resolved manifest for the reported run, it is not used to overwrite the paper’s 10-epoch protocol. One audited APR branch used AdamW, BF16, four GPUs, per-GPU batch size eight (global batch 32), learning rate 5×10^{-7} with minimum 2.5×10^{-7} , three epochs, advantage temperature 0.05, and weight clip 5. We identify those values as *branch-verified*, not as proof that every reported APR round used an identical schedule.

The table marks each row as reported, branch-verified, historical, or *proposed reproduction*. Where the final manifest is unresolved, the proposed schedule uses AdamW with $(\beta_1, \beta_2) = (0.9, 0.95)$, weight decay 10^{-4} , BF16, norm clipping at 1.0, and cosine decay. GT-only and PC-MTS use learning rate 10^{-4} , a three-epoch warmup, minimum learning rate 10^{-6} , eight scenes per GPU and two-step gradient accumulation on eight GPUs (global batch 128). FF-PGRPO uses 10^{-4} , two groups per GPU, four-step accumulation on eight GPUs (64 scene groups per update), the paper’s 16 rollouts per group, and the regularization schedule in Section A.2. APR follows the branch-verified optimizer settings and trains three epochs per promoted round. Proposed training seeds are 20260726, 20260727, and 20260728; they are future rerun identifiers and are not assigned to any existing point estimate.

No new multi-GPU training was started for this supplement. The launchers are dry-run by default and require both `ALLOW_TRAIN=1` and an explicit `-execute` flag. Wall-clock time and storage are explicitly recorded as *not measured*; no scheduler estimate is presented as an experiment. Offline cost is reported in evaluator calls rather than guessed wall time: one call per retained scene trajectory for a benchmark evaluation, eight additional calls per Stage-1 robustness center under the proposed tube, and at most four calls per APR teacher under the proposed interpolation list. Every future launch records wall time, peak device memory, output bytes, resolved configuration, Git revision, and SHA-256 hashes.

C.3 Benchmark Evaluation

NAVSIM v1. The scene-level reproduced export contains 12,146 single-trajectory predictions on `navtest`; 12,138 have metric caches and are scored. Their mean is PDMS 0.9145046521, with NC 0.9852941176, DAC 0.9798978415, TTC 0.9600428407, comfort 1.0, and EP 0.8674798204. These reproduce the rounded 91.4/98.5/98.0/96.0/100/86.7 row in the main paper. The archived inference seed is 20260726. The eight unscored predictions are excluded from component and aggregate means and remain listed in provenance; they are not imputed. The scorer implementation is mapped to `ba5908d:navsim/evaluate/pdm_score.py:pdm_score`.

NAVSIM v2. The reproduced `navtest` export has 12,146 successful trajectories and mean EPDMS 0.8911677436. Its component means reproduce the rounded main-paper row: NC 98.4, DAC 97.7, DDC 99.0, TLC 99.7, TTC 97.7, EP 89.4, LK 91.7, HC 97.1, and EC 87.7. The official evaluator revision recorded with this export is `0a380a9063d7162ec93d0f51e9990ebac585f720`. EC has 10,040 direct per-scene values and is aggregated by the official evaluator; we do not pretend that the other 2,106 rows contain directly observed EC entries.

Table 3: Stage-wise training protocol. Reported/reproduced settings are distinguished from bounded reproduction settings; ‘not recorded’ is an explicit provenance value, not a blank. Inference always emits one trajectory.

field	GT-only	PC-MTS	FF-PGRPO	APR	status
Target	logged GT	one sampled GT/candidate	sampled policy trajectories	audited teacher + policy retention	mixed provenance
Epochs	200	200	10	3 per audited branch; 3 paper rounds	paper reported / APR branch reproduced
Optimizer	AdamW	AdamW	AdamW	AdamW	historical implementation / reproduction setting
Peak learning rate	1e-4	1e-4	1e-4	5e-7	historical configs / APR reproduced
Minimum learning rate	1e-6	1e-6	1e-6	2.5e-7	historical configs / APR reproduced
Weight decay	1e-4	1e-4	1e-4	1e-4	historical / proposed for APR
Batch per GPU	8	8	2	8	historical / FF and APR manifests
GPUs	8	8	8	4	paper reports 8 overall; APR branch used 4
Gradient accumulation	2	2	4	1	historical / FF and APR manifests
Effective batch	128	128	64	32	deterministic from rows above
Precision	bf16-mixed	bf16-mixed	bf16-mixed	bf16-mixed	historical / APR reproduced
Warmup epochs	3	3	not recorded	0	historical / APR reproduced
Gradient clipping	1.0	1.0	1.0	1.0	historical / proposed where unresolved
Rollouts per scene	not applicable	16 support rollouts	16	8 offline candidates	proposed PC-MTS; paper FF; APR manifest
Advantage temperature	not applicable	not applicable	group normalized	0.05	APR reproduced
Advantage/weight clip	not applicable	not applicable	3.0	5.0	historical FF default / APR reproduced
BC or retention weight	not applicable	not applicable	0.10 to 0.05	1.0 retention	archived FF / APR reproduced
Reference KL	not applicable	not applicable	0.02	0.02 prior distance	archived FF / APR reproduced
Inference trajectories	1	1	1	1	paper and submission provenance

Table 4: Scene-level reproduction of the AMPT main results under one-trajectory inference without test-time scoring or reranking. All metrics are percentages and higher is better. NAVSIM v1 has 12,138 scored scenes; v2 has 12,146.

Benchmark	Scenes	Inference	Runs	NC	DAC	TTC	C	EP	PDMS	DDC	TLC	LK	HC	EC	EPDMS
NAVSIM v1	12138	1 trajectory	1 archived inference seed	98.53	97.99	96.00	100.00	86.75	91.45	not recorded	not recorded	not recorded	not recorded	not recorded	not recorded
NAVSIM v2	12146	1 trajectory	1 export; seed not recorded	98.42	97.69	97.66	not recorded	89.45	not recorded	99.00	99.71	91.73	97.09	87.74	89.12

Single-trajectory inference. For each scene and inference seed, the model receives one observation, runs one five-step denoising chain, and submits its single final 8×3 trajectory. There is no test-time group sampling, evaluator call, candidate scoring, reference lookup, model ensemble, or reranking. The v1 submission provenance independently records candidate count one, so this is a scene-level reproduced property rather than only a methodological statement.

Runs and uncertainty. The archived v1/v2 headline exports each establish a scene-level mean for one policy realization; they do not establish across-seed variance. We therefore do not attach a fabricated standard deviation to the 0.1–0.3 point increments in the paper. For the proposed rerun, each of seeds 20260726/20260727/20260728 produces a complete scene export. We first compute the mean for each run and report the mean $\pm$ standard deviation across the three runs. Method differences use 10,000 paired bootstrap resamples of shared scenes (resampling seeds and then scenes within seed when multiple seeds are present), with a fixed analysis seed of 2027 and a two-sided 95% interval. Recovery proportions use Wilson 95% intervals. Scene bootstrap intervals from one model run describe variation over the evaluated scenes and are explicitly not interpreted as training-seed stability.

Model selection disclosure. The archived final v1 provenance states that `navtest` participated in checkpoint selection. We therefore describe the headline table as a paper-reported benchmark comparison, not as a clean held-out model-selection study. The proposed rerun selects checkpoints only on the training/validation partition, seals a checkpoint hash before `navtest`, and evaluates each sealed hash once per inference seed. This disclosure does

not change a main paper number; it narrows the fairness claim to what the available provenance supports.

C.4 The Fixed 658-Scene Recovery Protocol

The hard population is fixed before comparing optimization methods. A scene enters the set when the ReCogDrive IL policy receives PDMS zero in all five sampling attempts because of collision, drivable-area violation, or both. The resulting intersection contains 658 scene IDs. A method *recovers* a scene when its single submitted trajectory has PDMS strictly greater than zero under the same evaluator; the threshold is not retuned per method.

For consistency with the main paper, every supplementary hard-set analysis uses the same archived scalar-GRPO versus historical Core-Pareto recovery pair that produced 367 and 440 positive-score scenes. It does not substitute a later full-benchmark checkpoint. On the 658 fixed scenes, scalar GRPO has 367 positive and 291 zero-score outcomes, whereas the manuscript recovery checkpoint has 440 positive and 218 zero-score outcomes. Their paired transition counts are 347 retained successes, 20 new failures, 93 repaired failures, and 198 persistent failures. Hence the net recovery is $93 - 20 = 73$ scenes, exactly the change $440 - 367$, rather than a gain obtained by hiding newly failed scenes. The corresponding recovery rates are 55.8% (Wilson 95% CI 52.0–59.5; $n = 658$) and 66.9% (63.2–70.4; $n = 658$). A 10,000-sample paired scene bootstrap gives an absolute difference of 11.1 percentage points with a 95% interval of 8.1–14.3 points. These intervals describe one archived checkpoint pair; no training-seed identifier was retained for this hard-set export.

The fixed-set construction contains 480 DAC-only, 165 collision-only, and 13 joint collision-and-DAC cases ac-

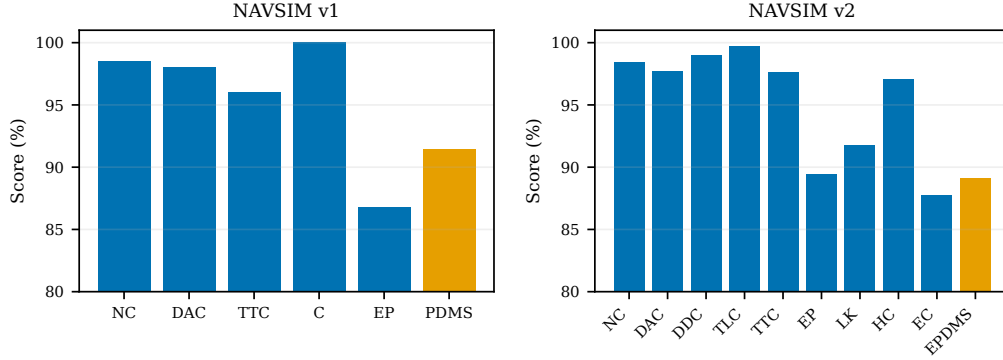


Figure 1: Scene-level reproduced AMPT component profiles on NAVSIM v1 and v2. All metrics are percentages and higher is better; aggregate PDMS/EPDMS is shown in orange. The profiles use 12,138 and 12,146 scored scenes, respectively, from one inference realization.

cording to the archived cause manifest. The same IDs and failure labels are used in all hard-set transition and case-table analyses. In particular, “440 recovered” means 440 positive outcomes on the fixed original population; the transition table is required whenever the claim is interpreted as a policy-to-policy improvement.

C.5 Baseline and Controlled-Comparison Protocol

AMPT’s v1 and v2 rows are locally reproduced from scene-level outputs. Other headline baseline rows are values reported in the main paper and their cited sources; no local checkpoint, common prediction export, or common seed manifest was found for them. They are consequently labeled *reported*, not *reproduced*. The table records backbone, sensor interface, training data, number of inference candidates, and use of a test-time scorer whenever the cited evidence establishes those fields. An unavailable field is marked “not established by the audited source” rather than filled by analogy with AMPT.

For the GT/Score/Pareto/PC-MTS comparison, the paper reports a shared initialization, diffusion SFT loss, optimizer, epoch budget, scalar-GRPO configuration, and evaluation protocol. The controlled reproduction additionally seals the candidate-source manifest, fixes the proposed non-GT sampling probability, matches the accepted candidate count, draws one target per scene-update, and uses the same three proposed training seeds. For Scalar GRPO versus FF-PGRPO, all rollout chains are generated from the same initialization and sampled with the same seed before the credit function is changed. For No APR versus Full APR, total optimizer steps are matched by continuing the no-APR policy with retention targets. These are proposed control procedures until their launch manifests and outputs exist; the main-paper point estimates remain labeled paper-reported.

C.6 Reproduction and Integrity Checks

The canonical dataset stores normalized scores, benchmark and split, method, stage, variant, round, scene ID, seed when recorded, feasibility/failure transitions, teacher status, and its source-file/key provenance. Validation checks expected scene counts, duplicate scene-method records, score ranges, the benchmark aggregate formulas, paired scene sets, and failure-state logic. Final tables and figures are generated from that dataset; a validation failure stops manuscript generation. The build manifest records the command, analysis seed, package versions, Git revision, and SHA-256 hashes of inputs and outputs. All raw result directories are read-only, and all derived files remain under the supplementary directory.

D PC-MTS Analyses: Supervision–Optimization Alignment

This section is needed because the quality of a supervision target and its value for subsequent policy optimization are different quantities. PC-MTS is designed for the latter: it keeps a target only when its score, component-wise trade-off, local feasibility, and proximity to the frozen learner’s rollout support are jointly acceptable. We first examine the observed imitation-to-optimization reversal, and then specify the controls required to separate policy compatibility from candidate count and training budget.

D.1 The observed supervision–optimization reversal

Table 6 restates the reported comparison from the main paper and makes the change induced by the common scalar-GRPO stage explicit. Score and Pareto selection produce the two strongest imitation checkpoints (87.2 and 87.5 PDMS), yet both deteriorate under the subsequent

Table 5: Fair-comparison protocol for main-paper baselines. All methods use one inference candidate and no test-time scorer in the paper table; unreported backbone/sensor/data details are not inferred. Result identity distinguishes reported from reproduced.

method	backbone	sensors	training_data	inference_candidates	test_time_scorer	result_identity
AMPT	InternVL3-2B + ReCogDrive diffusion planner	multi-view in paper; one audited APR command resolves single camera	NAVSIM plus offline candidate/teacher pools	1	no	locally reproduced v1/v2 final rows; camera-manifest discrepancy disclosed
ReCogDrive	not disclosed in current PDF	not disclosed in current PDF	not disclosed in current PDF	1	no	paper reported
DiffusionDriveV2	not disclosed in current PDF	not disclosed in current PDF	not disclosed in current PDF	1	no	paper reported
Drive-JEPA	not disclosed in current PDF	not disclosed in current PDF	not disclosed in current PDF	1	no	paper reported

Table 6: Supervision-optimization comparison on NAVSIM v1 under single-trajectory inference. All four rows are paper-reported point estimates whose run/seed aggregation is not stated; higher PDMS and GRPO change are better, and no variance is implied.

method	il_pdms	post_scalar_grpo_pdms	grpo_change_points	evidence	runs
GT	86.4	90.3	3.9	paper reported	reported point; run/seed aggregation not stated
Score	87.2	85.8	-1.4	paper reported	reported point; run/seed aggregation not stated
Pareto	87.5	86.7	-0.8	paper reported	reported point; run/seed aggregation not stated
PC-MTS	86.9	91.1	4.2	paper reported	reported point; run/seed aggregation not stated

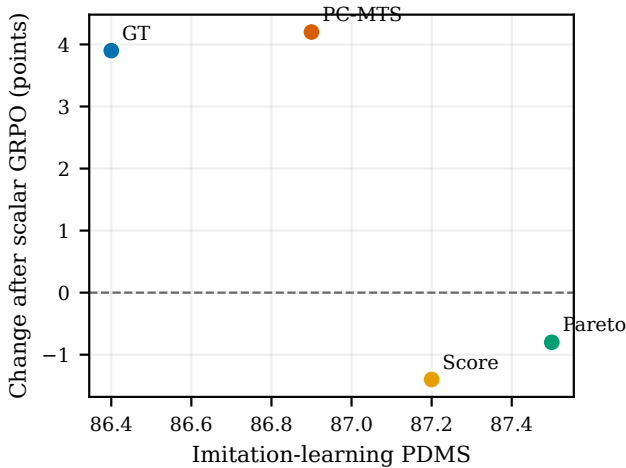


Figure 2: Paper-reported imitation PDMS versus the change after the common scalar-GRPO stage on NAVSIM v1. A high imitation score alone does not predict optimization value: Score and Pareto regress, whereas PC-MTS gains 4.2 points. Each marker is one reported point estimate; the paper does not state run/seed aggregation and no synthetic error bars are shown.

optimizer. In contrast, PC-MTS deliberately gives up 0.3–0.6 imitation points relative to those selectors and improves by 4.2 points after the same optimization stage. GT supervision also improves, but finishes 0.8 points below PC-MTS. Thus, ranking target sets by imitation PDMS would select Pareto, whereas ranking them by optimized PDMS selects PC-MTS. This reversal is the direct evidence for the mismatch claimed in the paper.

The comparison should be interpreted at the level supported by the evidence. It demonstrates that a higher-scoring imitation checkpoint need not be a better starting distribution for GRPO, and that the PC-MTS set is compatible with a large positive update in this run. The archived paper table does not contain scene-level outputs, acceptance counts, or multiple seeds for these four cells. Consequently, it does not by itself establish that the reversal is invariant to seed, nor does it isolate candidate

count from the four filtering decisions. We therefore do not attach synthetic error bars or policy-distance values to these points.

D.2 Candidate-count-matched reconstruction protocol

The released reproduction configuration turns the missing causal controls into a predeclared, low-ambiguity experiment. All four variants start from the same GT-only checkpoint, use the same candidate sources, contribute exactly one target per scene and update, train imitation learning for 200 epochs, and then run scalar GRPO with 16 rollouts per scene for 10 epochs. Within each navigation-command/source stratum, accepted candidates are deterministically subsampled to the smallest retained count across variants. The non-GT sampling probability is held fixed, so a method cannot receive more gradient updates merely because it retains more candidates. Empty strata fall back to the logged trajectory. In addition to final PDMS, a 300-update diagnostic records the early GRPO gain before long-horizon training effects can dominate.

For a fully specified reconstruction when the original resolved PC-MTS manifest is unavailable, the release marks the following as *proposed reproduction settings*, not measured settings of the reported run: a rollout bank of 16, $K = 3$ neighbors, a command-conditional 0.95 calibration quantile, and eight temporally correlated local perturbations. The rollout bank and calibration queries are disjoint by construction. Three predeclared training seeds (20260726, 20260727, and 20260728) are used for the reconstruction, while bootstrap seed 2027 is reserved for analysis and is never treated as an additional training run. These choices fill the executable configuration without retroactively attributing unrecorded hyperparameters to the main-paper result.

For each run, the pipeline records retained candidates, acceptance rate, candidate-source composition, feasible rollout rate, feasible high-quality rate, all-infeasible-group rate, zero-score rate, the 10th percentile, CVaR-10, 300-update gain, and final scalar-GRPO PDMS. Method differences are computed on identical scenes with a 10,000-resample paired bootstrap; means and standard deviations are reported across the three training seeds.

D.3 Filter attribution and sensitivity

The component experiment is cumulative: (i) aggregate Score only, (ii) Score plus the component-wise Pareto

gate, (iii) the preceding gates plus calibrated policy compatibility, and (iv) the full selector with local-feasibility testing. Counts are matched after each gate, rather than relaxing a stricter gate to reach a target count. This distinction matters: a matched count must not reintroduce candidates that failed the mechanism being tested. The decisive diagnostic is not imitation PDMS alone, but whether adding a gate raises feasible rollout density and converts the fixed-budget GRPO change from negative to positive.

The companion sensitivity protocol varies one factor at a time: rollout-bank size in $\{8, 16, 32\}$, K in $\{1, 3, 5\}$, calibration quantile in $\{0.90, 0.95, 0.975\}$, and perturbation count in $\{4, 8, 16\}$. Compatibility alternatives are compared at the same acceptance rate: candidate-to-GT ADE/FDE, distance to the deterministic policy output, minimum rollout-bank distance, average K -nearest distance, and command-calibrated K -nearest distance. These are evidence-bounded planned analyses: the release contains their configurations and logging schema, but no unexecuted outcome is used to support the paper’s reported 91.1 PDMS.

Taken together, the existing reversal and the matched protocol close two different parts of the argument. The former establishes that isolated target quality can disagree with downstream optimization value; the latter defines the experiment needed to attribute that disagreement specifically to policy compatibility rather than supervision quantity.

E FF-PGRPO Analyses: Feasibility Before Positive Credit

Aggregate reward can hide an unsafe trade: progress or comfort may increase enough to give positive standardized advantage even when collision avoidance, drivable-area compliance, or a protected reference metric regresses. The purpose of FF-PGRPO is therefore not merely to reshape the scalar score, but to change which rollouts are eligible for positive credit.

E.1 What the stage ablation establishes

On NAVSIM v1, the no-stage baseline in the main ablation has 86.5 PDMS, 98.1 NC, 94.7 DAC, 94.2 TTC, and 80.9 EP. Applying FF-PGRPO without PC-MTS reaches 91.0 PDMS, with 98.4 NC, 98.0 DAC, 95.8 TTC, and 85.93 EP. With PC-MTS as the initialization, FF-PGRPO reaches 91.1 PDMS, 98.5 NC, 98.0 DAC, 96.0 TTC, and 86.0 EP before APR. The improvement is therefore not produced by sacrificing the displayed hard-safety components for progress: DAC improves by 3.3 points and TTC by 1.6 points in the FF-PGRPO-only row, while EP increases by 5.03 points. These are paper-reported point estimates whose run/seed aggregation is not stated, so the small 0.1-point difference between the two FF-PGRPO rows is

not presented as a stable gain.

This result is consistent with the proposed ordering—recover feasibility, then optimize continuous quality—but an aggregate stage ablation cannot directly reveal the sign of individual rollout credits. The fixed hard-scene transition in Section G.1 provides scene-level evidence, and the instrumentation below defines the rollout-level test.

E.2 Credit diagnostics

For rollout i , let $A_i > 0$ denote positive assigned advantage and let $V_i = 1$ denote a violation of any hard-feasibility, protected-reference, or admissible-Pareto condition. We report

$$\text{PCVR} = \frac{\sum_i \mathbb{I}[A_i > 0] V_i}{\sum_i \mathbb{I}[A_i > 0]}, \quad (5)$$

and separately count reference-regressing and Pareto-dominated positive credits. A denominator of zero is reported as “no positive credits,” not as zero violation. FF-PGRPO’s implementation makes a violating rollout ineligible for positive credit and exposes assertions for this contract. The archived training run, however, does not contain the per-rollout trace needed to quote an observed PCVR; we do not convert a code invariant into a fabricated measured rate.

The stepwise mechanism ablation adds: (i) the hard feasibility gate, (ii) the coherent-reference guard, (iii) the Pareto positive-credit gate, and (iv) the all-infeasible recovery rule. Every variant logs group state (all-feasible/mixed/all-infeasible), feasibility rate, positive and negative advantage means, PCVR, reference-regressing positive-credit rate, dominated-positive-credit rate, zero-to-feasible repair, positive-to-zero regression, and updates until the first feasible rollout. The coherent reference is always one executable trajectory. The historical component-wise envelope remains only as an ablation because combining the best value of each metric from different trajectories may describe no realizable plan and can make every sampled rollout appear to regress.

The low-cost diagnostic run uses the paper’s group size of 16 and stops at 300 updates; the final run uses the reported 10 epochs. All variants share the same sampled rollout tensors when credit is recomputed offline, which makes the initial credit-assignment comparison independent of policy drift. New training is required only for the subsequent learning curves and remains disabled unless explicitly enabled.

E.3 Hard-scene evidence for feasibility-first recovery

The manuscript recovery checkpoint is evaluated on the same 658-scene set as scalar GRPO. Scalar GRPO produces a positive score in 367 scenes (55.8%, 95% Wilson CI 52.0–59.5%), whereas the AMPT recovery checkpoint is positive in 440 scenes (66.9%, CI 63.2–70.4%). The

Table 7: Stage-wise NAVSIM v1 ablation under single-trajectory inference. Higher is better. Rows are paper-reported point estimates whose run/seed aggregation is not stated; only the final AMPT row is independently reproduced from 12,138 scene scores.

configuration	pcmts	ff_pgrpo	apr	NC	DAC	TTC	C	EP	PDMS	evidence	runs
GT-only	no	no	no	98.1	94.7	94.2	100.0	80.9	86.5	paper reported	reported point; run/seed aggregation not stated
PC-MTS	yes	no	no	98.2	95.2	94.5	100.0	81.3	86.9	paper reported	reported point; run/seed aggregation not stated
FF-PGRPO only	no	yes	no	98.4	98.0	95.8	100.0	85.93	91.0	paper reported	reported point; run/seed aggregation not stated
PC-MTS + FF-PGRPO	yes	yes	no	98.5	98.0	96.0	100.0	86.0	91.1	paper reported	reported point; run/seed aggregation not stated
Full AMPT	yes	yes	yes	98.5	98.0	96.0	100.0	86.7	91.4	paper reported / final row scene-reproduced	reported point + 12138 final scenes; run aggregation not stated

comparison uses a single trajectory per scene and neither method uses test-time sampling, scoring, or reranking. Crucially, the paired transition is not merely a difference of gross counts: 93 scalar-GRPO zeros become positive, 20 scalar-GRPO positives become zero, 347 successes are retained, and 198 failures persist. Net recovery is therefore $93 - 20 = 73$ scenes, or 11.1 percentage points. A 10,000-resample scene-paired bootstrap (analysis seed 2027) gives a 95% CI of 8.1–14.3 percentage points for this paired change. This interval quantifies scene variation for this fixed checkpoint pair; it does not replace missing multi-seed training uncertainty.

The subset mean PDMS increases from 0.5101 to 0.6145. The corresponding component means change from 0.8381 to 0.8815 for NC, 0.7052 to 0.7781 for DAC, 0.7705 to 0.8161 for TTC, and 0.4956 to 0.5932 for EP; comfort increases from 0.9985 to 1.0000, while DDC changes from 0.9567 to 0.9521. This last decrease is why protected metrics must be reported next to aggregate recovery rather than summarized as universal component-wise improvement. The complete transition and failure-type decomposition are given in Section G.1.

The aggregate ablation, paired hard-scene transition, and positive-credit instrumentation serve complementary roles. The first shows the overall stage gain, the second verifies that recovery is net positive rather than purchased entirely with new failures, and the third is the predeclared test of the claimed credit-assignment mechanism.

F APR Analyses: Dynamic, Policy-Relative Teachers

APR addresses a moving-target problem. A trajectory that is a useful local teacher for the current policy can become redundant after refinement, while a previously distant candidate can enter the policy’s compatible neighborhood. The relevant question is therefore not whether additional optimization steps help, but whether rebuilding and re-auditing the teacher set provides useful targets while retaining already safe behavior.

F.1 Round-wise behavior and protected components

The main paper reports a monotonic sequence of 91.10, 91.21, 91.37, and 91.45 PDMS from Round 0 through Round 3, corresponding to gains of +0.11, +0.16, and

Table 8: Paper-reported NAVSIM v1 APR progression under single-trajectory inference. Higher is better. Run/seed aggregation is not stated and no training-seed confidence interval is available.

round	pdms	gain_points	evidence	runs
0	91.1	0.0	paper reported	reported point; run/seed aggregation not stated
1	91.21	0.11	paper reported	reported point; run/seed aggregation not stated
2	91.37	0.16	paper reported	reported point; run/seed aggregation not stated
3	91.45	0.08	paper reported / final score scene-reproduced	reported point + 12138 final scenes; run aggregation not stated

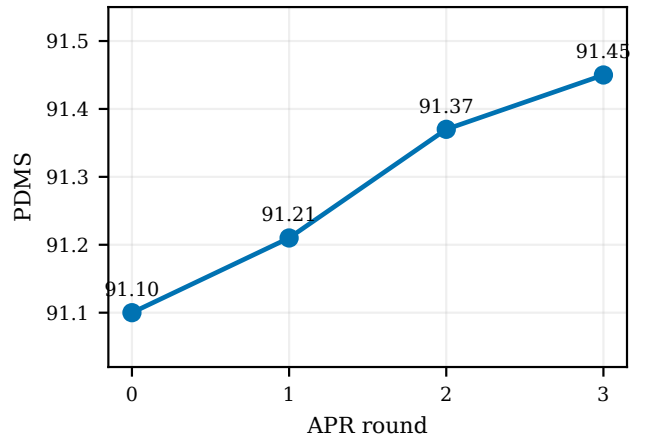


Figure 3: Paper-reported NAVSIM v1 progression across APR rounds. The final point is reproduced at scene level, but the small round increments are from one reported sequence and carry no across-seed error bar.

+0.08. In the stage ablation, adding APR to the PC-MTS+FF-PGRPO policy raises PDMS from 91.1 to 91.4 and EP from 86.0 to 86.7, while the displayed NC, DAC, TTC, and comfort values remain 98.5, 98.0, 96.0, and 100, respectively. This pattern is aligned with APR’s design: local teacher updates increase progress without visible regression in the protected components at the reported precision.

Each round value is a reported checkpoint summary; its run/seed aggregation is not stated. No across-seed standard deviation is available, and the 0.08–0.16 point increments are not claimed to be independently significant. Moreover, the archived experiment history contains accepted and rejected branches rather than a single complete three-round hash chain; the four values above are therefore treated as the paper’s round-level summary, not as permission to infer missing intermediate measurements.

Table 9: Audited APR teacher-pool construction on NAVSIM navtrain. Counts are from one archived teacher audit; the 28,910 accepted teachers follow interpolation and exact re-evaluation. Higher acceptance is not inherently better because guards constrain quality.

step	source	count	acceptance_definition	evidence
source selection	DeiVor	25402	source-specific strict candidates	archived APR runbook
source selection	DiffusionDriveV2	13859	source-specific strict candidates	archived APR runbook
strict union	merged sources	39261	feasible and component-protected before interpolation	deterministic sum of audited source counts
post-interpolation audit	accepted teachers	28910	alpha 0.25 then exact re-evaluation	archived APR runbook

F.2 Teacher audit and post-interpolation evaluation

One traceable APR teacher-construction pass starts from 39,261 strict external trajectories: 25,402 (64.7%) from one driving-policy source and 13,859 (35.3%) from a second source. After interpolation with coefficient 0.25 and exact re-evaluation, 28,910 targets remain, an acceptance rate of 73.6%; the remaining 10,351 are rejected. These counts are from one audited navtrain construction pass and are not presented as the source proportions of every APR round. They nevertheless establish an important operational fact: interpolation is followed by evaluator-based admission, and the check is non-vacuous—more than one quarter of the pre-audit union does not enter the distillation set.

Teacher status is defined relative to the current policy. A candidate is *accepted* when its evaluated local target is feasible, exceeds the current single prediction by the required margin, preserves protected metrics, and stays inside the current rollout neighborhood. An accepted teacher becomes *expired* if it fails those tests after the policy is updated. A previously rejected candidate is *newly activated* if it passes after the rollout bank and current prediction are regenerated. These definitions prevent lifecycle counts from being conflated with raw pool size.

The interpolation search is descending: larger moves are evaluated first and the first admissible move is retained. Every accepted target stores the teacher, current prediction, interpolation coefficient, post-interpolation score, protected-metric deltas, compatibility distance, and evaluator hash. No-interpolation and no-post-evaluation controls remove different safeguards: the former jumps directly to the teacher, whereas the latter retains a local trajectory without verifying that interpolation preserved feasibility.

F.3 Step-matched controls

The release defines eight variants from the same Round-0 checkpoint and matches the total number of optimizer updates: extra training on the existing mixture, one-shot distillation, repeated use of a fixed teacher set, dynamic teachers, dynamic teachers without interpolation, without post-interpolation re-evaluation, without policy retention, and full APR. The dynamic-teacher and full-APR variants rebuild predictions, rollout banks, and teacher audits at the same three round boundaries reported in the paper. The fixed-teacher variant repeats the Round-1 set without re-ranking it; the extra-training control receives

Table 10: APR fixed-teacher diagnostic on NAVSIM v1. The archived fixed-set continuation decreases from its 91.2766 anchor, whereas the 91.45 endpoint is the paper’s compressed dynamic-round report; the rows are diagnostic rather than a step-matched causal ablation.

policy_or_epoch	pdms	interpretation	evidence
dynamic-teacher anchor	91.2766	accepted dynamic-teacher checkpoint	archived APR runbook
fixed-set continuation epoch 0	91.2526	below dynamic-teacher anchor	archived APR runbook
fixed-set continuation epoch 1	91.2311	further below dynamic-teacher anchor	archived APR runbook
paper APR round 3	91.45	compressed round-level paper report	paper reported

the same update count but no new teacher target. This design separates four possible explanations: more updates, teacher supervision in general, teacher-set rebuilding, and the interpolation/retention safeguards.

For every round, the required report contains PDMS and all components, regressed scenes, new failures, accepted/expired/newly activated teachers, mean and median verified teacher advantage, interpolation-coefficient distribution, and scene coverage. A checkpoint is promoted only if its single-trajectory aggregate score improves and protected metrics pass the same non-regression audit; otherwise the previous checkpoint remains the dynamic anchor. Training ends after three promoted rounds or when no audited local target remains. These controls are configured but not represented as measured results in this supplement, because running them would require new training.

Optional task-vector consolidation is kept separate from teacher discovery. The implementation can mask low-salience coordinates, require directional agreement, bound each branch norm and the total displacement from the current anchor, and then evaluate the merged checkpoint. The best-branch, simple average, task-arithmetic, and TIES-style controls are required before assigning an independent gain to merging [1, 6]. In the absence of a stable, step-matched gain, merging should be read as a conservative checkpoint-consolidation detail, not as the source of APR’s teacher lifecycle.

The present evidence therefore supports two bounded statements: the reported APR rounds are monotonic and preserve the displayed safety/compliance metrics, and an audited construction pass materially filters proposed local targets after re-evaluation. The step-matched variants above are the necessary test for the stronger causal claim that dynamic rebuilding, rather than additional training alone, explains the full increment.

G Failure Recovery and Qualitative Diagnostics

Qualitative evidence is most useful when organized by a mechanism and tied to scene-level metrics. We therefore analyze the fixed hard set first, then give traceable examples of recovered, persistent, and newly introduced failures. All quantitative hard-set results and the mechanism-organized case table use the same historical checkpoint pair as the main paper’s 367-versus-440 recovery claim.

Table 11: Recovery on the same 658 NAVSIM v1 scenes under one archived comparison. Rates use Wilson 95% intervals over scenes; this interval does not represent training-seed uncertainty. Higher recovery is better.

Method	Positive	Zero	Recovery rate (%)	Wilson 95% CI (%)	Runs	Scenes
Scalar GRPO	367	291	55.8	[52.0, 59.5]	1 archived comparison	658
AMPT recovery checkpoint	440	218	66.9	[63.2, 70.4]	1 archived comparison	658

Table 12: Paired recovery change on 658 NAVSIM v1 scenes. The 11.1-point gain has a scene-paired bootstrap interval (10,000 resamples, analysis seed 2027); higher net repair is better. One archived model pair is available.

Absolute gain (pp)	Paired 95% CI (pp)	Zero-to-positive	Positive-to-zero	Net repair	Scenes	Runs
11.1	[8.1, 14.3]	93	20	73	658	1 archived comparison

The final full-benchmark visualization is separately labeled and is not used in the 658-scene recovery analysis.

G.1 Construction of the fixed 658-scene set

The set contains 658 NAVSIM v1 scenes for which the initial ReCogDrive policy obtained exactly zero PDMS in all five construction samples because of an at-fault collision (NC), drivable-area violation (DAC), or both. The set is fixed before comparing the scalar-GRPO and AMPT recovery checkpoints. Under single-trajectory evaluation, scalar GRPO is positive on 367 and zero on 291 scenes; the AMPT recovery checkpoint is positive on 440 and zero on 218.

The paired 2×2 transition is

scalar GRPO	AMPT recovery checkpoint		row total
	zero	positive	
zero	198	93	291
positive	20	347	367
column total	218	440	658

(6)

Hence 93 failures are repaired, 347 successes are retained, 198 failures persist, and 20 new failures appear. The net change is $93 - 20 = 73$, exactly the difference between the gross positive counts $440 - 367$. This distinction is important: $440/658$ is the gross recovery rate relative to the policy used to construct the hard set, whereas 93 and 20 describe the paired transition from scalar GRPO to the AMPT recovery checkpoint. The resulting 11.1-percentage-point paired improvement has a scene-bootstrap 95% CI of 8.1–14.3 points (10,000 resamples, seed 2027). It is a net improvement rather than an exchange of an equal number of old and new failures.

The original hard-set taxonomy contains 480 DAC-only scenes, 165 collision-only scenes, 13 scenes failing both, and no residual “other” category. The paired repairs/new failures are 59/14 for DAC-only, 32/6 for collision-only, and 2/0 for the joint category, giving net changes of 45, 26, and 2 scenes. The gains are therefore not concentrated in only one failure type, although the small joint category is descriptive rather than statistically decisive.

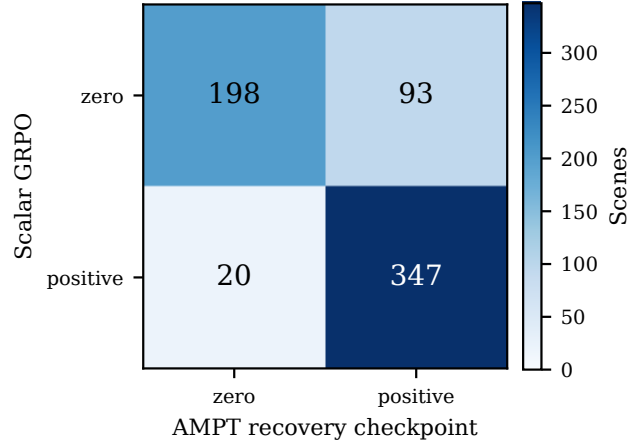


Figure 4: Scalar-GRPO to AMPT transition on the fixed 658-scene set. The off-diagonal cells expose both 93 repairs and 20 regressions, rather than reporting gross recovery alone.

Table 13: Failure-type decomposition for the fixed 658 NAVSIM v1 scenes. Repairs and regressions use the scalar-GRPO to AMPT transition; net repairs sum to 73. One archived comparison; higher net repair is better.

Failure type	Scenes	Repairs	Regressions	Net repair
DAC-only/all-fail	480	59	14	45
collision-only/all-fail	165	32	6	26
collision + DAC	13	2	0	2

G.2 Traceable scene cases

Table 14 selects cases by transition type, not by visual appeal. Navigation commands come from the corresponding NAVSIM scene metadata; scores and components come from the fixed paired evaluation. The table uses full scene identifiers so every row can be regenerated. Component pairs are scalar-GRPO $\rightarrow$ AMPT-recovery values.

The first two cases are consistent with feasibility-oriented recovery: a hard component changes from zero to one and positive PDMS returns. Scene-level before/after metrics do not identify the temporal order of training updates. The third shows why “recovered” does not mean component-wise perfect. The fourth remains a DAC/progress failure under both audited checkpoints; without a candidate/rollout manifest, the metrics do not identify whether pool coverage, policy support, or optimization is responsible. The fifth prevents survivorship bias in the qualitative analysis and motivates APR retention; it is one of the 20 positive-to-zero transitions already charged against net recovery.

G.3 Mechanism-centered visualization protocol

The supplied renderer groups panels by decision rather than by scene similarity. A PC-MTS panel overlays an accepted candidate, a rejected distant candidate, the

Table 14: Mechanism-organized examples from the fixed 658-scene set. Scene IDs and PDMS values are reproduced from the archived comparison; commands are joined from audited scene metadata. These selected cases are not an unbiased benchmark sample.

Scene ID	Command	Outcome	PDMS	NC	DAC	TTC	EP	DDC	Diagnosis
00fcd6d092c5e8e	left	repaired	0.000 -> 1.000	0.000 -> 1.000	1.000 -> 1.000	0.000 -> 1.000	0.000 -> 1.000	1.000 -> 1.000	Collision/TTC failure repaired without protected regression.
03aa8a0576a25b63	right	repaired	0.000 -> 0.963	1.000 -> 1.000	0.000 -> 1.000	1.000 -> 1.000	0.000 -> 0.912	1.000 -> 1.000	Drivable-area failure repaired while NC and TTC remain feasible.
1148c72f141c532d	left	partial repair	0.000 -> 0.583	0.000 -> 1.000	0.000 -> 1.000	0.000 -> 0.000	0.000 -> 1.000	1.000 -> 1.000	Both hard failures repaired; TTC remains the limiting component.
00016f8b45c25a1d	left	persistent	0.000 -> 0.000	1.000 -> 1.000	0.000 -> 0.000	1.000 -> 1.000	0.000 -> 0.000	0.000 -> 0.000	DAC/progress failure persists under both audited checkpoints.
4b4a268bee4c5ab5	left	regressed	1.000 -> 0.000	1.000 -> 1.000	1.000 -> 0.000	1.000 -> 1.000	1.000 -> 0.000	1.000 -> 1.000	A DAC regression creates a new zero and motivates retention.

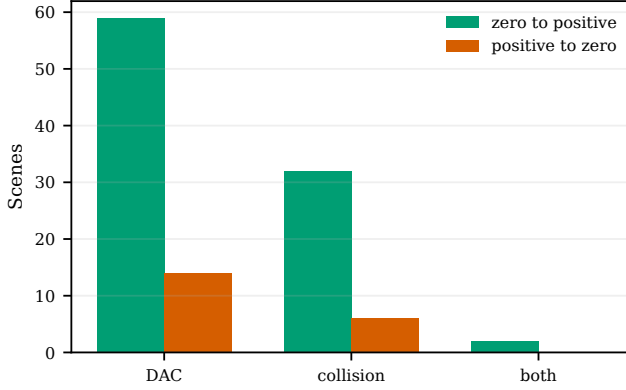


Figure 5: Repairs and regressions by the fixed hard-set cause taxonomy. The net changes are 45 DAC, 26 collision, and 2 joint cases, totaling 73.

logged trajectory, and the frozen-policy rollout neighborhood. An FF-PGRPO mixed-group panel distinguishes unsafe, feasible-but-dominated, and admissible Pareto-positive rollouts and prints their assigned advantages; an all-infeasible panel shows the coherent recovery reference. An APR panel shows the current prediction, raw teacher, evaluated interpolant, and the accepted/expired/newly-activated lifecycle state. Every rendered panel reads the scene id, command, component scores, and method name from its manifest and uses a common legend: blue for the current prediction, green for an accepted or positively credited trajectory, orange for a dominated/expired trajectory, red for an unsafe trajectory, and black dashed for the coherent reference.

This protocol is also a safeguard against overinterpretation. A camera/BEV image is emitted only when the frame, command, all trajectories, and evaluator record share the same scene identifier. The metric-only cases in Table 14 remain quantitative case studies if those visual assets are unavailable; no unrelated camera frame or hand-drawn trajectory is substituted merely to complete a panel.



Figure 6: Three selected full-benchmark NAVSIM v1 cases (left, straight, and right commands) rendered from the separately audited final-AMPT manifest. Each row contains the complete front frame, bird's-eye view, scene ID, command, color legend, PDMS, and trajectory error. These examples are not members of, and contribute no statistics to, the historical 658-scene recovery comparison; they illustrate final-policy behavior only.

H Additional Related Work, Fair Comparison, and Limitations

H.1 Related perspectives

Support-aware policy improvement. Conservative of-line policy-improvement methods limit updates in regions where the behavior data provide weak support. SPIBB constrains uncertain decisions to the baseline policy [3], while BEAR regularizes a learned policy toward the behavior distribution [2]. PC-MTS shares the support intuition but applies it before optimization: it filters trajectory-level supervision using samples from the frozen learner, with command-conditional calibration and a local feasibility test. It is not a general monotonic-improvement guarantee; its role is to avoid moving the imitation policy into a region where downstream rollout optimization has poor local support.

Multi-objective and Pareto optimization. Multi-objective reinforcement learning studies policies under vector-valued returns and alternative preference models [5]. AMPT does not seek a global set of Pareto-optimal policies. PC-MTS uses a scene-local Pareto filter for supervision, and FF-PGRPO uses a hierarchy: hard feasibility and coherent-reference guards are evaluated before a rollout can receive positive Pareto credit. This order is essential when an aggregate score can compensate for a protected regression.

Advantage-weighted behavior learning. Advantage-weighted regression turns estimated advantages into weights for supervised policy updates [4]. APR is related at the loss level, but its weights are attached only after a candidate is compared with the current single-trajectory prediction, locally interpolated, and re-evaluated. Teacher admission and policy retention are therefore as important as the weighting function itself.

Model merging. Task arithmetic combines model displacements relative to a shared anchor [1], and TIES resolves redundancy and sign conflicts before merging [6]. AMPT may use an anchor-constrained variant to consolidate APR branches, but merging is optional and is separated from teacher selection. Passing parameter-space norm and sign checks does not imply behavioral safety; every merged checkpoint must still undergo the same single-trajectory evaluator audit.

H.2 Fair-comparison boundary

AMPT uses one predicted trajectory at inference. Policy rollout banks, candidate pools, coherent references, group sampling, specialist policies, and the offline evaluator are training-only; there is no test-time scorer, best-of- N selection, or candidate reranking. This contract is present in the paper and is independently supported by the v1 submission provenance.

The released v1 scene export contains 12,138 success-

fully scored scenes from 12,146 predictions; the eight unscored predictions are disclosed rather than imputed. The v2 export contains 12,146 successful scene scores. Its EC aggregation follows the official evaluator, although a direct EC value is present for 10,040 rows. Scene-bootstrap intervals resample these scored scenes and are explicitly labeled as scene uncertainty; they are not a substitute for independent training runs.

The baseline rows in the main tables are treated as *reported* unless a local scene export and matching protocol are available. AMPT’s v1/v2 rows are reproduced from scene-level outputs, but protocol matching is not established for every external baseline. Moreover, the audited v1 provenance records use of navtest for checkpoint selection. We therefore avoid a strict held-out state-of-the-art claim and report protocol fields—backbone, sensors, training data, inference candidates, and test-time scoring—alongside each comparison.

The 658-scene result uses the recovery checkpoint and checkpoint pair specified by the main-paper analysis. It is a fixed-subset diagnostic, not a claim that every later checkpoint must induce the same transitions. Reporting the full 2×2 matrix in Eq. 6 makes the gross and net notions of recovery explicit.

H.3 Limitations

Finite support estimates. Compatibility is estimated from a finite rollout bank. Sparse commands or multimodal scenes can make a genuinely useful target appear distant, and the calibrated neighborhood is only as representative as the frozen policy samples. Increasing bank size improves coverage but raises offline inference and storage cost.

Evaluator bias and non-reactive evaluation. All three stages use an offline NAVSIM evaluator for filtering, credit, or teacher validation. Systematic evaluator errors can therefore be amplified across stages. NAVSIM’s non-reactive agents also cannot reveal how other road users would respond to the learned trajectory; feasibility under this scorer is not a closed-loop safety guarantee.

Candidate-pool ceiling. APR can select and localize only behaviors represented in its multi-source pool. If no source proposes a valid maneuver, rebuilding the pool changes no target. The persistent case in Table 14 establishes persistence under two audited checkpoints, but cannot diagnose this pool ceiling without a per-scene candidate manifest.

Locality can reject long-horizon value. The policy-compatibility filter intentionally favors learnable local targets. It may reject a distant behavior that is initially difficult to imitate but would be valuable after a longer curriculum. Policy-dependent teacher-value and cross-policy selection experiments are therefore important follow-ups.

Benchmark-specific hierarchy. The ordering of NC/DAC feasibility, DDC/TLC guards, progress floors, and continuous Pareto objectives reflects NAVSIM met-

rics. Transferring AMPT to a new benchmark requires revalidating the hierarchy and calibration thresholds, not merely renaming metric fields.

No behavioral semantics for parameter merging. Task-vector masks, direction agreement, and norm constraints bound parameter movement but do not provide a semantic or safety guarantee. If its independent gain is not stable across seeds and step-matched controls, merging should remain an implementation detail.

Statistical scope. The supervision comparison, stage ablation, and APR round table are available as reported point estimates rather than three-seed aggregates. Small APR increments must not be described as stable until the configured multi-seed controls are run. The paired 658-scene interval establishes a robust scene-level net transition for one checkpoint pair, but does not measure training-seed variation.

These limitations narrow rather than alter the paper’s central claim. The currently traceable evidence establishes the supervision-optimization reversal, a large feasibility-oriented stage gain, monotonic reported APR rounds with preserved displayed safety components, and positive net recovery on a fixed hard set. Candidate-count matching, rollout-level credit traces, step-matched APR controls, and multi-seed reruns are the remaining experiments needed for stronger causal and stability claims.

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